

Credit Card Fraud Detection

Logistic Regression | Python | scikit-learn

284,807

Transactions

492

Fraud Cases

95.43%

Test Accuracy

30

Features (PCA)

The Dataset

284,807 European credit card transactions from Sept 2013

FEATURE

DESCRIPTION

Time

Seconds elapsed since first transaction

V1 – V28

PCA-transformed anonymized features (28 cols)

Amount

Transaction value in USD

Class

0 = Legitimate | 1 = Fraudulent

Zero null values across all 284,807 rows — Clean & ready for ML

The Big Problem:

Class Imbalance

99.83%

Legitimate

● 284,315

Legitimate (99.83%)

● 492

Fraudulent (0.17%)

Naive model predicting ALL Legit = 99.83% accuracy – completely useless!

We need a smarter approach to handle this severe imbalance.

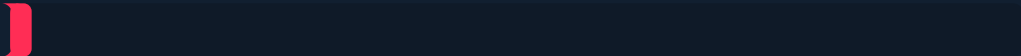
Solving Imbalance: Under-Sampling

BEFORE

Legit 284,315 samples



Fraud 492 samples



AFTER (Balanced)

Legit 492 samples



Fraud 492 samples



Python Code:

```
legit = dataset[dataset.Class == 0]
fraud = dataset[dataset.Class == 1]
legit_sample = legit.sample(n=492)
new_dataset = pd.concat([legit_sample, fraud])
```

Result: 984 balanced samples | 50% Legit | 50% Fraud

```
pd.concat([legit_sample, Fraud], axis=0)
```

The ML Pipeline



FEATURES USED	TARGET VARIABLE	TRAIN SIZE	TEST SIZE
V1-V28 (PCA) + Amount + Time	Class [0=Legit, 1=Fraud]	787 samples (80% of 984)	197 samples (20% of 984)

Model Performance

787 training samples
TRAINING ACCURACY

94.41%

`model.score(x_train, y_train)`

197 test samples
TESTING ACCURACY

95.43%

`model.score(x_test, y_test)`

TRAIN



94.41%

TEST



95.43%

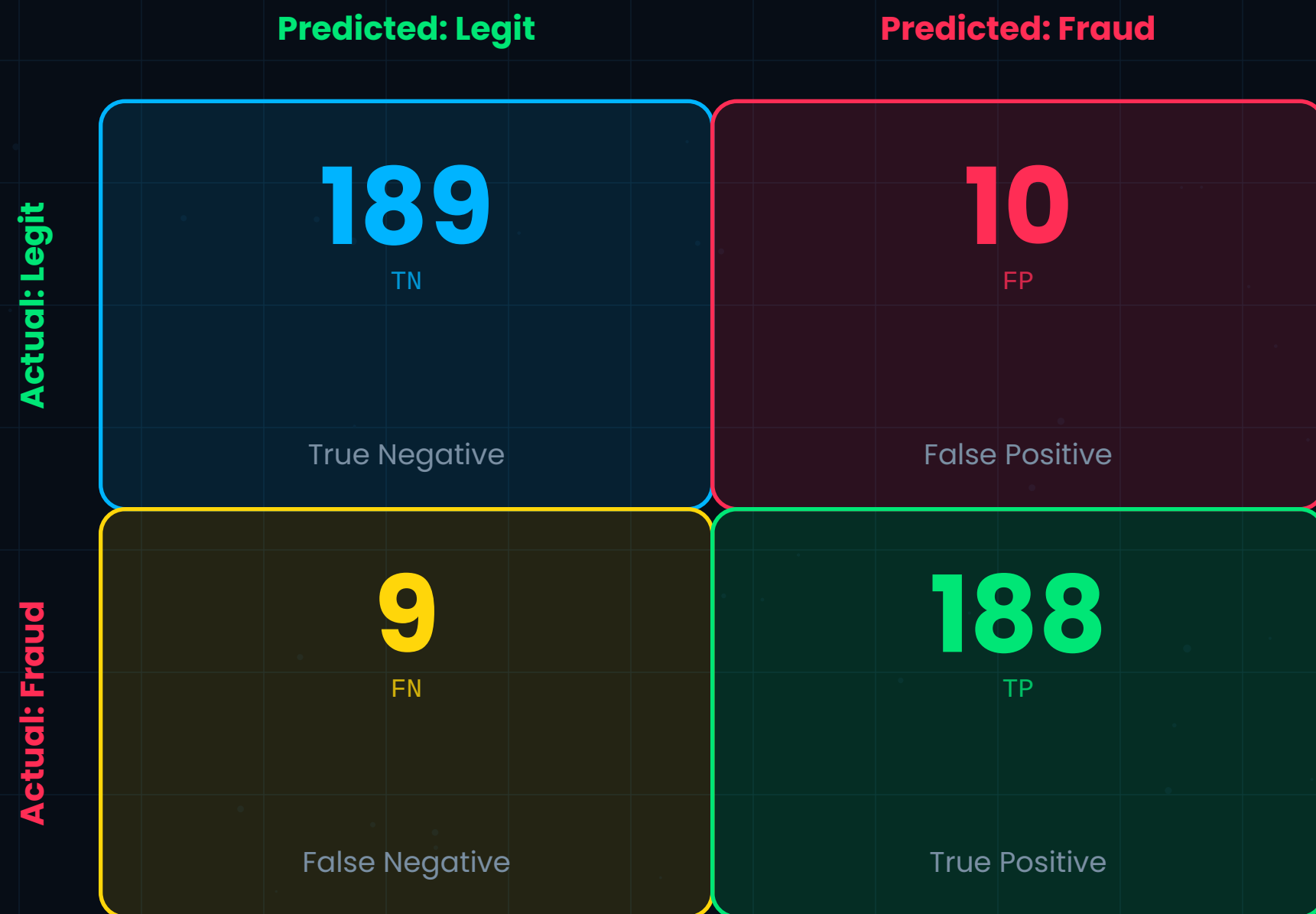
Test Accuracy > Train Accuracy No Overfitting Detected!

The model generalizes well to unseen data

```
model = LogisticRegression()
model.fit(x_train, y_train)
print(model.score(x_train, y_train)*100) # → 94.41
print(model.score(x_test, y_test)*100)  # → 95.43
```

Confusion Matrix

Breaking down model predictions on test set (~197 samples)



PRECISION

94.9%

RECALL

95.4%

F1-SCORE

95.1%

ACCURACY

95.43%

Feature Importance

Top PCA features — Mean value difference: Fraud vs Legit class



V3 (-7.03) and V7 (-5.57) are the strongest fraud signals in this dataset

Legit transactions have mean ≈ 0 for all PCA features; fraud deviates significantly

Key Insights

No Overfitting

Test accuracy (95.43%) exceeds training accuracy (94.41%), confirming the model generalizes well to unseen transactions.

Imbalance is the Real Challenge

0.17% fraud rate means naive models fail. Under-sampling creates a fair 50/50 split for meaningful training.

Fraud Amount is Bimodal

Fraud mean (\$122.21) > Legit mean (\$88.29), BUT fraud median (\$9.25) < legit median (\$22.00) — small probes + large strikes.

PCA Features Reveal Patterns

V3 (-7.03) and V7 (-5.57) deviate massively from zero in fraud, making them the strongest discriminative features for detection.

Data Quality is Perfect

Zero null values across all 31 columns and 284,807 rows. No imputation needed — clean pipeline from load to model.

Tech Stack Used

pandas

Data loading, EDA,
groupby & concat

numpy

Numerical ops &
array handling

scikit-learn

LogisticRegression
train_test_split

matplotlib

Visualization &
EDA plots

Jupyter

Interactive
notebook env

Project Summary

Dataset: **284,807 rows | 31 features | 0 nulls**

Model: **Logistic Regression | Under-sampling strategy**

Result: **94.41% Train | 95.43% Test | No overfitting**

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